

System Design Document

The system is designed to support real-time property price prediction through a web interface. Users provide property attributes, which are validated and sent to the backend prediction service.

Functional requirements include property selection, prediction generation, error handling, and result display. Non-functional requirements include performance, reliability, maintainability, and scalability.

The model training workflow includes data ingestion, preprocessing, label encoding, train-test splitting, model training, evaluation, and model persistence. Prediction requests follow a structured workflow from frontend to backend and finally to the machine learning model.

The design ensures modularity by separating frontend, backend, preprocessing, and training components.